

```
In [2]: import pandas as pd
df = pd.read_csv('dataset.csv')
```

```
In [3]: df.head()
```

```
Out[3]:
```

	Unnamed: 0	DATE	Month	Year	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME_1	Brand	PACK_SIZE
0	0	2018-10-17	October	2018	1	1000	1	5	Natural Chip Company Chip Compny SeaSalt175g	Natural Chip Company	175g
1	1	2019-05-14	May	2019	1	1307	348	66	CCs Nacho Cheese 175g	CCs	175g
2	2	2019-05-20	May	2019	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	Smiths	170g
3	3	2018-08-17	August	2018	2	2373	974	69	Smiths Chip Thinly S/Cream&Onion 175g	Smiths	175g
4	4	2018-08-18	August	2018	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g	Kettle	150g

```
In [4]: df['DATE']=pd.to_datetime(df['DATE'])
```

```
In [5]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264835 entries, 0 to 264834
Data columns (total 15 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Unnamed: 0            264835 non-null int64
 1   DATE                  264835 non-null datetime64[ns]
 2   Month                 264835 non-null object
 3   Year                  264835 non-null int64
 4   STORE_NBR             264835 non-null int64
 5   LYLTY_CARD_NBR        264835 non-null int64
 6   TXN_ID                264835 non-null int64
 7   PROD_NBR              264835 non-null int64
 8   PROD_NAME_1           264835 non-null object
 9   Brand                 264835 non-null object
10   PACK_SIZE             264835 non-null object
11   PROD_QTY              264835 non-null int64
12   TOT_SALES             264835 non-null float64
13   LIFESTAGE             264835 non-null object
14   PREMIUM_CUSTOMER      264835 non-null object
dtypes: datetime64[ns](1), float64(1), int64(7), object(6)
memory usage: 30.3+ MB

```

```

In [25]: tot=df['TOT_SALES'].sum()
         print(f'{tot:,.2f}')

```

1,805,171.70

```

In [6]: df['PROD_NAME_1'].head(20)

```

```

Out[6]: 0      Natural Chip Company Chip Compny SeaSalt175g
        1              CCs Nacho Cheese 175g
        2      Smiths Crinkle Cut Chips Chicken 170g
        3      Smiths Chip Thinly S/Cream&Onion 175g
        4      Kettle Tortilla ChpsHny&Jlpno Chili 150g
        5      Old El Paso El Paso Salsa Dip Tomato Mild 300g
        6      Smiths Crinkle Chips Salt & Vinegar 330g
        7      Grain Waves Waves Sweet Chilli 210g
        8      Doritos Corn Chip Mexican Jalapeno 150g
        9      Grain Waves Waves Sour Cream&Chives 210G
        10     Smiths Crinkle Chips Salt & Vinegar 330g
        11     Kettle Sensations Siracha Lime 150g
        12              Twisties Cheese 270g
        13     Woolworths Crinkle Cut Chicken 175g
        14              Thins Chips Light& Tangy 175g
        15              CCs Original 175g
        16              Burger Rings 220g
        17      Natural Chip Company Sour Cream & Garden Chive...
        18      Doritos Corn Chip Southern Chicken 150g
        19              Cheezels Cheese Box 125g
        Name: PROD_NAME_1, dtype: object

```

```

In [7]: words=(df['PROD_NAME_1'].drop_duplicates().str.split( ).explode())
        words

```

```

Out[7]: 0      Natural
        0      Chip
        0      Company
        0      Chip
        0      Compny
        ...
        549     150g
        689     Doritos
        689     Salsa
        689     Mild
        689     300g
        Name: PROD_NAME_1, Length: 617, dtype: object

```

```

In [8]: product_words = words.to_frame(name='words')

```

```
In [9]: product_words = product_words[product_words['words'].str.fullmatch(r'[A-Za-z]+')]
product_words
```

```
Out[9]:
```

	words
0	Natural
0	Chip
0	Company
0	Chip
0	Compny
...	...
549	Cut
549	Bolognese
689	Doritos
689	Salsa
689	Mild

465 rows × 1 columns

```
In [10]: product_words.groupby(['words'])['words'].count().sort_values(ascending=False)
```

```
Out[10]: words
      Chips      21
      Smiths     16
      Crinkle    14
      Chip       14
      Cut        14
      ..
      Vinegr      1
      Whlegrn     1
      Vingar      1
      Whlgrn      1
      saltd       1
      Name: words, Length: 164, dtype: int64
```

```
In [11]: product_words['words'].unique()
```

```
Out[11]: array(['Natural', 'Chip', 'Company', 'Compny', 'CCs', 'Nacho', 'Cheese',
      'Smiths', 'Crinkle', 'Cut', 'Chips', 'Chicken', 'Thinly', 'Kettle',
      'Tortilla', 'Chili', 'Old', 'El', 'Paso', 'Salsa', 'Dip', 'Tomato',
      'Mild', 'Salt', 'Vinegar', 'Grain', 'Waves', 'Sweet', 'Chilli',
      'Doritos', 'Corn', 'Mexican', 'Jalapeno', 'Sour', 'Sensations',
      'Siracha', 'Lime', 'Twisties', 'Woolworths', 'Thins', 'Tangy',
      'Original', 'Burger', 'Rings', 'Cream', 'Garden', 'Chives',
      'Southern', 'Cheezels', 'Box', 'Infuzions', 'Crn', 'Crnchers',
      'Gcamole', 'Sea', 'And', 'Red', 'Rock', 'Deli', 'Thai', 'Pringles',
      'Sthrn', 'FriedChicken', 'BBQ', 'SR', 'Mzzrlla', 'Originl',
      'saltd', 'Sp', 'Truffle', 'Swt', 'Mexicana', 'French', 'OnionDip',
      'ChipCo', 'Hony', 'Soy', 'Chp', 'Supreme', 'Roast', 'Mozzarella',
      'Basil', 'Pesto', 'SweetChili', 'PotatoMix', 'Camembert', 'Fig',
      'smiths', 'Mac', 'N', 'Honey', 'Seasonedchicken', 'Rib', 'Prawn',
      'Crackers', 'Plus', 'Btroot', 'Jam', 'Tyrrells', 'Crisps',
      'Lightly', 'Salted', 'Medium', 'Pot', 'SourCream', 'Onion',
      'Chnky', 'Tom', 'Cobs', 'Popd', 'Co', 'Tmato', 'Fries', 'Potato',
      'Med', 'RRD', 'Coconut', 'Hot', 'Spicy', 'Crm', 'Crnkle', 'Orgnl',
      'Big', 'Bag', 'Crips', 'Stacked', 'Tostitos', 'Barbecue',
      'Cheetos', 'Puffs', 'Splash', 'Of', 'Tasty', 'Smoked', 'Chipotle',
      'Barbeque', 'Mystery', 'Flavour', 'Ched', 'Snbts', 'Whlgrn', 'Chs',
      'Bacon', 'Balls', 'Slt', 'Vingar', 'Veg', 'Strws', 'Mango',
      'Chutny', 'Papadums', 'Steak', 'Chimuchurri', 'Sunbites',
      'Whlegrn', 'Pepper', 'Vinegr', 'Aioli', 'Slow', 'Rst', 'Pork',
      'Belly', 'Pc', 'Bolognese'], dtype=object)
```

```
In [12]: df=df[~df['PROD_NAME_1'].str.contains("SALSA",case=False)]
```

```
In [13]: df.head()
```

```
Out[13]:
```

	Unnamed: 0	DATE	Month	Year	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME_1	Brand	PACK_SIZE
0	0	2018-10-17	October	2018	1	1000	1	5	Natural Chip Company Chip Compny SeaSalt175g	Natural Chip Company	175g
1	1	2019-05-14	May	2019	1	1307	348	66	CCs Nacho Cheese 175g	CCs	175g
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3	3	2018-08-17	August	2018	2	2373	974	69	Smiths Chip Thinly S/Cream&Onion 175g	Smiths	175g
4	4	2018-08-18	August	2018	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g	Kettle	150g

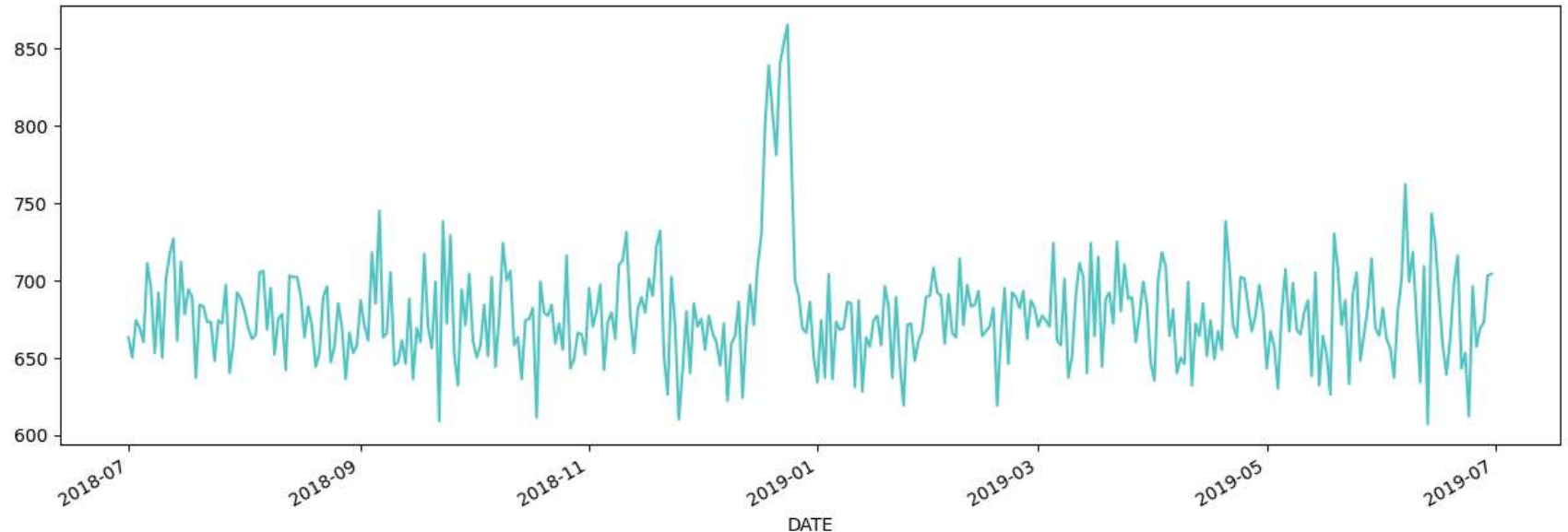
```
In [14]: df.groupby(['LYLTY_CARD_NBR'])['PROD_QTY'].sum().sort_values(ascending=False)
```

```
Out[14]: LYLTY_CARD_NBR
226000    400
230078     36
162039     34
179228     32
116181     32
...
1024        1
1023        1
1022        1
1019        1
1009        1
Name: PROD_QTY, Length: 71288, dtype: int64
```

```
In [15]: df=df[df['LYLTY_CARD_NBR']!= 226000 ]
```

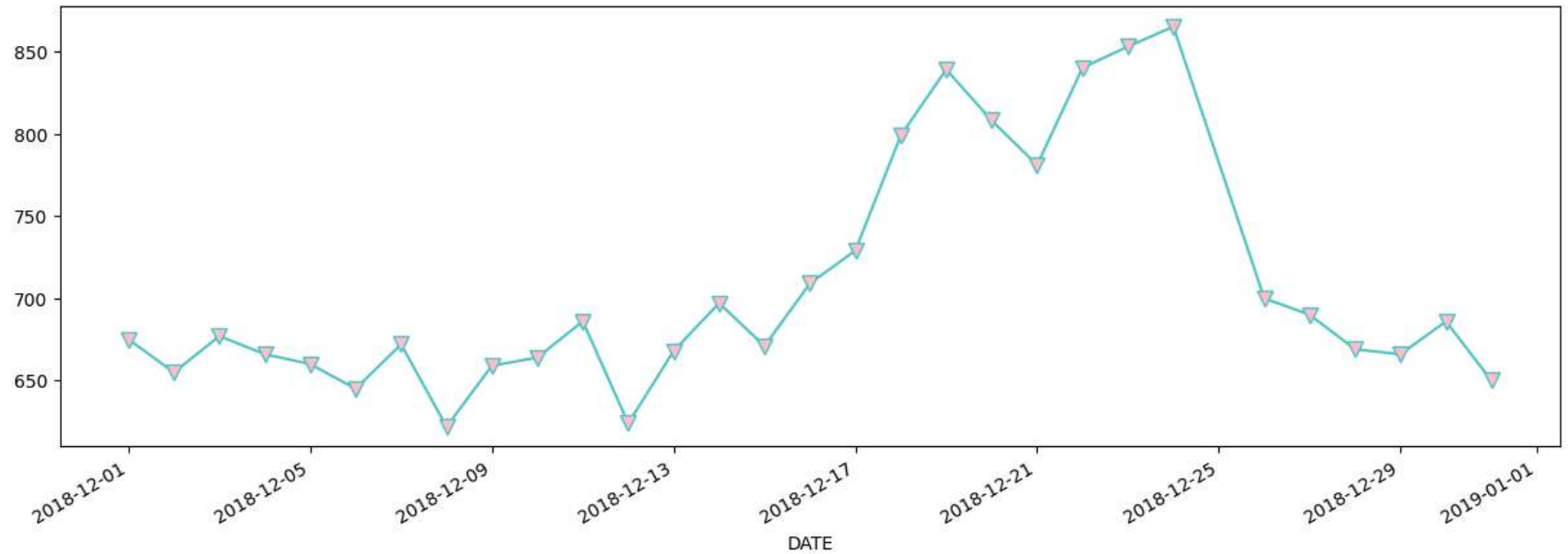
```
In [36]: import matplotlib.pyplot as plt
df.groupby(['DATE'])['TXN_ID'].count().plot(kind='line',figsize=(15,5),color = '#50c5c7')
```

```
Out[36]: <Axes: xlabel='DATE'>
```



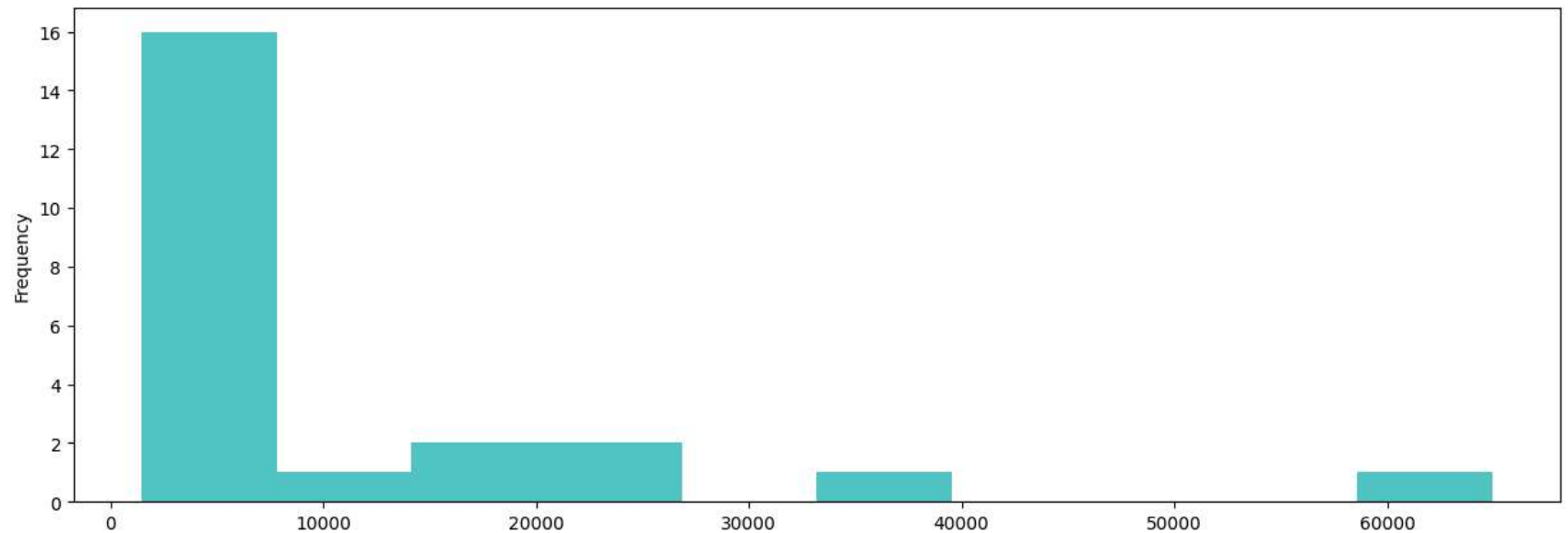
```
In [17]: dec=df[df['DATE'].dt.month==12]
dec.groupby(df['DATE'])['TXN_ID'].count().plot(kind='line',figsize=(15,5),color = '#50c5c7',marker="v",markersize=9,r
```

Out[17]: <Axes: xlabel='DATE'>



```
In [18]: df.groupby(df['PACK_SIZE'])['TXN_ID'].count().plot(kind='hist',figsize=(15,5),color = '#50c5c7')
```

Out[18]: <Axes: ylabel='Frequency'>

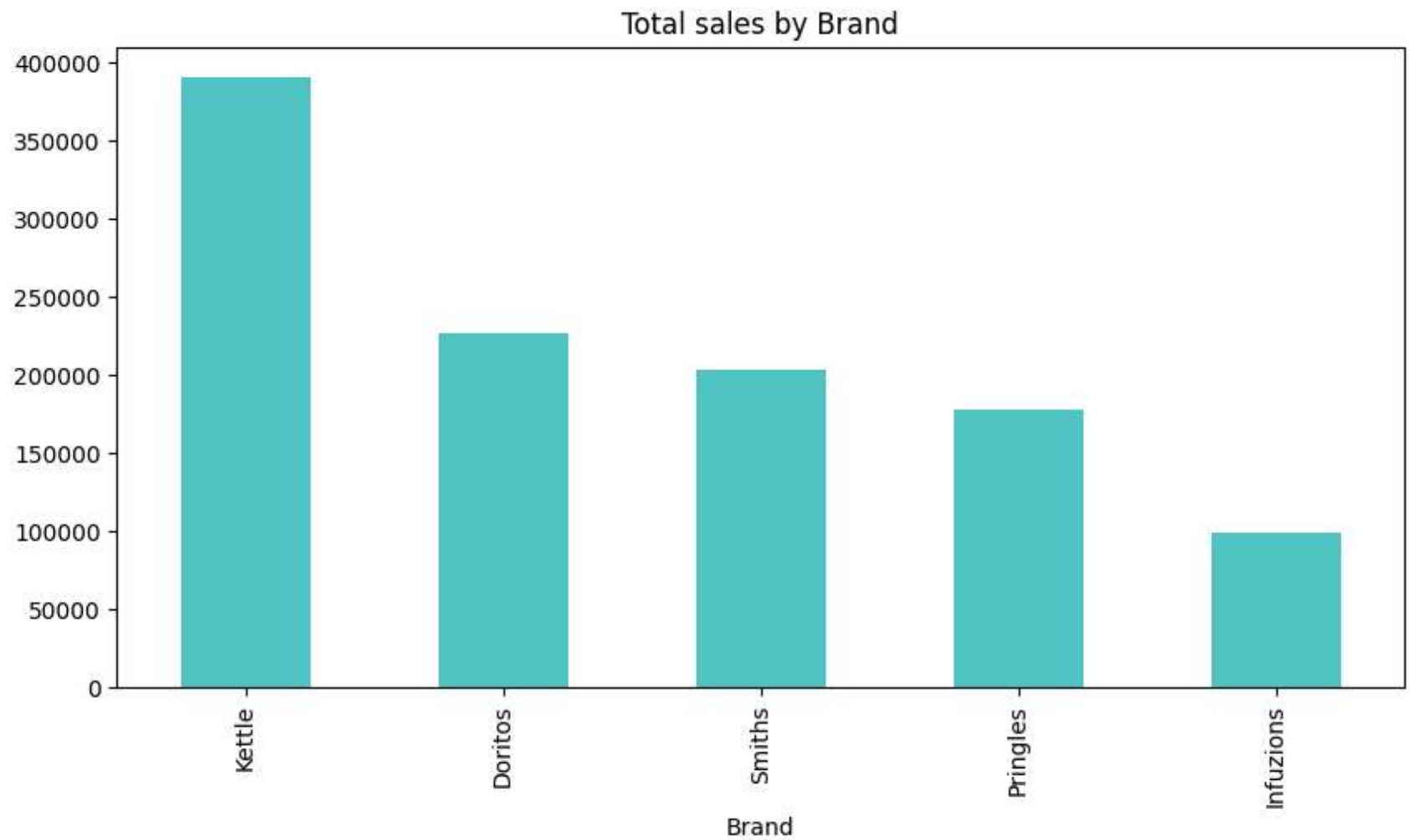


```
In [19]: print(df.groupby(df['Brand'])['TOT_SALES'].sum().sort_values(ascending = False).apply(lambda x : f"₹{x:,.2f}"))
```

```
Brand
Kettle                ₹390,239.80
Doritos               ₹226,329.90
Smiths               ₹202,902.60
Pringles             ₹177,655.50
Infuzions            ₹99,047.60
Thins                ₹88,852.50
Twisties             ₹81,522.10
Tostitos             ₹79,789.60
Cobs                 ₹70,569.80
RRD                  ₹64,954.50
Tyrrells             ₹51,647.40
Grain Waves          ₹43,044.20
Natural Chip Company ₹42,318.00
Woolworths           ₹41,059.10
Cheezels             ₹40,029.90
Red Rock Deli        ₹22,653.00
CCs                  ₹18,078.90
Cheetos              ₹16,884.50
smiths               ₹14,583.40
Grain Waves Waves    ₹8,546.70
French               ₹7,929.00
Burger               ₹6,831.00
Snbts                ₹5,076.20
Sunbites             ₹4,600.20
Grain Waves          ₹26.30
Name: TOT_SALES, dtype: object
```

```
In [20]: df.groupby(df['Brand'])['TOT_SALES'].sum().sort_values(ascending=False).head(5).plot(kind='bar',title = 'Total sales
```

```
Out[20]: <Axes: title={'center': 'Total sales by Brand'}, xlabel='Brand'>
```



```
In [21]: df.groupby(df['PREMIUM_CUSTOMER'])['TOT_SALES'].sum().sort_values(ascending=False).apply(lambda x : f"₹{x:,.2f}")
```

```
Out[21]: PREMIUM_CUSTOMER
Mainstream    ₹700,865.40
Budget        ₹631,406.85
Premium       ₹472,899.45
Name: TOT_SALES, dtype: object
```

```
In [22]: df.groupby(df['PACK_SIZE'])['TOT_SALES'].sum().sort_values(ascending = False).apply(lambda x : f"₹{x:,.2f}")
```

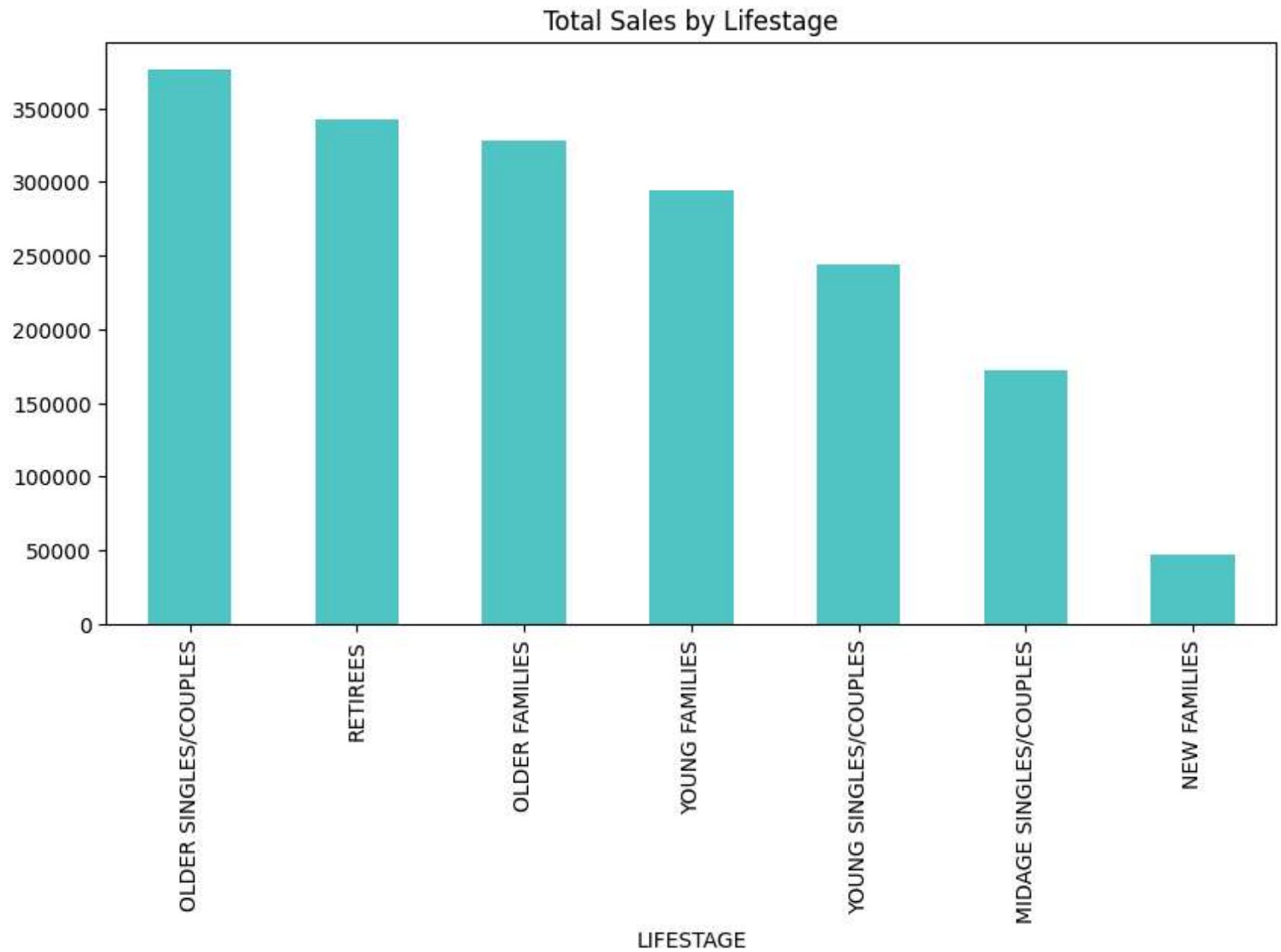
Out[22]: PACK_SIZE

175g	₹477,106.40
150g	₹282,003.00
134g	₹177,655.50
110g	₹162,765.40
170g	₹146,673.00
330g	₹136,794.30
165g	₹101,360.60
380g	₹75,419.60
270g	₹55,425.40
250g	₹26,096.70
Salt	₹26,090.40
210g	₹21,700.80
210G	₹21,348.00
200g	₹16,007.50
190g	₹14,412.90
160g	₹10,647.60
90g	₹9,676.40
180g	₹8,568.40
175G	₹8,325.00
150G	₹7,678.80
70g	₹6,852.00
220g	₹6,831.00
125g	₹5,733.00

Name: TOT_SALES, dtype: object

In [23]: `df.groupby(df['LIFESTAGE'])['TOT_SALES'].sum().sort_values(ascending=False).plot(kind='bar',title='Total Sales by Li'`

Out[23]: `<Axes: title={'center': 'Total Sales by Lifestage'}, xlabel='LIFESTAGE'>`



```
In [26]: avg_unit=(df.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER']).agg(PROD_QTY=('PROD_QTY', 'sum'),  
                                                                    unique_cus = ('LYLTY_CARD_NBR', 'nunique')))
```

```
avg_unit['avg_unit_cus']=(avg_unit['PROD_QTY'] / avg_unit['unique_cus'])
```

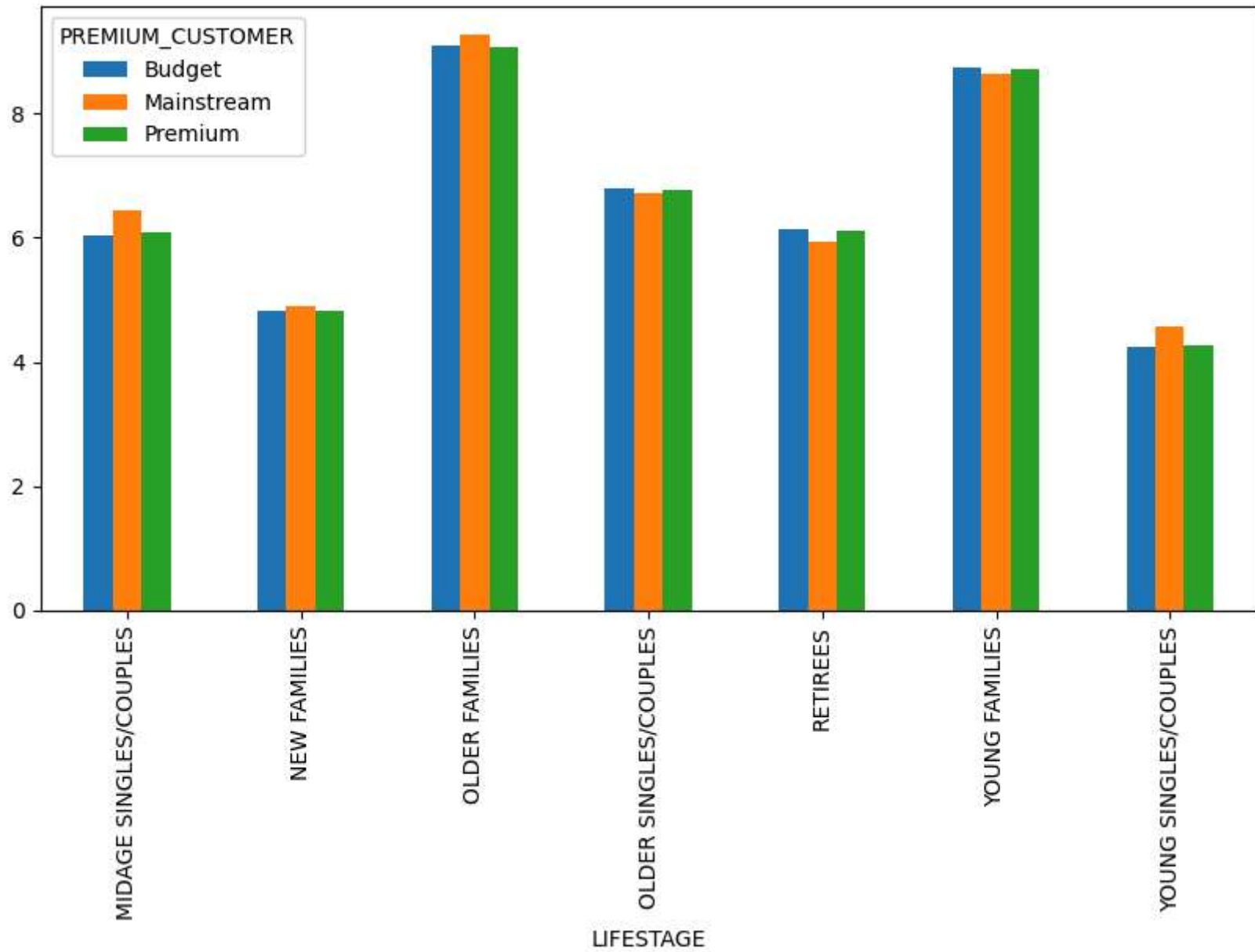
```
avg_unit
```

Out[26]:

		PROD_QTY	unique_cus	avg_unit_cus
LIFESTAGE	PREMIUM_CUSTOMER			
MIDAGE SINGLES/COUPLES	Budget	8883	1474	6.026459
	Mainstream	21213	3298	6.432080
	Premium	14400	2369	6.078514
NEW FAMILIES	Budget	5241	1087	4.821527
	Mainstream	4060	830	4.891566
	Premium	2769	575	4.815652
OLDER FAMILIES	Budget	41853	4611	9.076773
	Mainstream	25804	2788	9.255380
	Premium	20239	2231	9.071717
OLDER SINGLES/COUPLES	Budget	32883	4849	6.781398
	Mainstream	32607	4858	6.712021
	Premium	31693	4682	6.769116
RETIREEES	Budget	26932	4385	6.141847
	Mainstream	37677	6358	5.925920
	Premium	23266	3812	6.103358
YOUNG FAMILIES	Budget	34482	3953	8.722995
	Mainstream	23194	2685	8.638361
	Premium	20901	2398	8.716013
YOUNG SINGLES/COUPLES	Budget	15500	3647	4.250069
	Mainstream	36225	7917	4.575597
	Premium	10575	2480	4.264113

```
In [27]: avg_unit['avg_unit_cus'].unstack().plot(kind='bar',figsize=(10,5))
```

```
Out[27]: <Axes: xlabel='LIFESTAGE'>
```



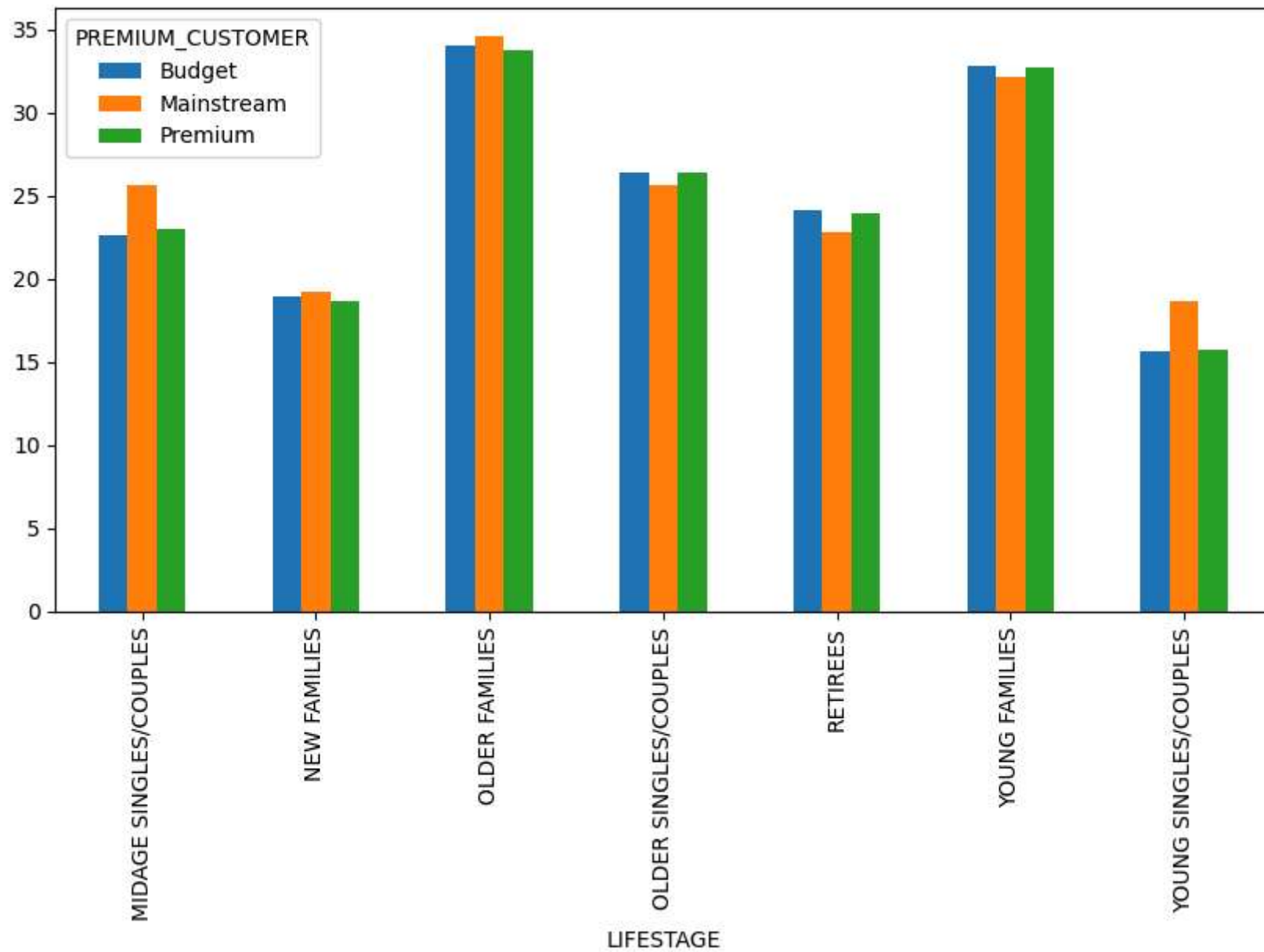
```
In [28]: avg_price=(df.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER']).agg(total_sales=('TOT_SALES', 'sum'),  
                                                                    unique_cus = ('LYLTY_CARD_NBR', 'nunique'))  
  
avg_price['avg_price_cus']=(avg_price['total_sales'] / avg_price['unique_cus'])  
  
avg_price
```

Out[28]:

		total_sales	unique_cus	avg_price_cus
LIFESTAGE	PREMIUM_CUSTOMER			
MIDAGE SINGLES/COUPLES	Budget	33345.70	1474	22.622592
	Mainstream	84734.25	3298	25.692617
	Premium	54443.85	2369	22.981786
NEW FAMILIES	Budget	20607.45	1087	18.958096
	Mainstream	15979.70	830	19.252651
	Premium	10760.80	575	18.714435
OLDER FAMILIES	Budget	156863.75	4611	34.019464
	Mainstream	96413.55	2788	34.581618
	Premium	75242.60	2231	33.725952
OLDER SINGLES/COUPLES	Budget	127833.60	4849	26.362879
	Mainstream	124648.50	4858	25.658399
	Premium	123531.55	4682	26.384355
RETIREEES	Budget	105916.30	4385	24.154230
	Mainstream	145168.95	6358	22.832487
	Premium	91296.65	3812	23.949803
YOUNG FAMILIES	Budget	129717.95	3953	32.815065
	Mainstream	86338.25	2685	32.155773
	Premium	78571.70	2398	32.765513
YOUNG SINGLES/COUPLES	Budget	57122.10	3647	15.662764
	Mainstream	147582.20	7917	18.641177
	Premium	39052.30	2480	15.746895

```
In [29]: avg_price['avg_price_cus'].unstack().plot(kind='bar',figsize=(10,5))
```

```
Out[29]: <Axes: xlabel='LIFESTAGE'>
```



SUMMARY

TOTAL_CHIP_SALES = ₹1,805,171.70

Highly saled Pack_sizes = 175g, 150g, 134g

Frequent Customer = Older singles/couples

High revenue Month = December

-> The results suggest that sales are not evenly distributed across all products and customer groups. -> A small number of brands, customer segments, and pack sizes appear to drive a large share of revenue. -> This indicates that focused marketing and inventory planning could improve business performance. -> Prioritize high-performing brands and expand promotional support around them.